

BRACKETS

Minutes 0–15: No cost. The airline's schedule buffer absorbs it. The pdf also states that under 15 is no delay

Minutes 15–60: Only marginal operating costs (crew time, fuel, maintenance) - the data from [16 Cost of delay – EUROCONTROL Standard Inputs for Economic Analyses](#) is used depending on the size of the aircraft to yield cost per minute

- *60 being the top bracket is also backed up by the typical narrowbody turnaround time (same boundary used in the model) beyond this, the cascade to the next leg becomes nearly certain.*

Minutes 60–120: Reactionary delays to other flights start cascading.

At minute 120 EU261 Article 9 kicks in for short-haul flights, triggering mandatory meal and refreshment vouchers

Minutes 120–180: EU261 Article 9 kicks in : meal/refreshment vouchers become mandatory. *the most operationally dangerous bracket because the dispatcher is racing the clock toward the cliff at 180 minutes where it is obligatory to pay out large compensation*

Minutes 180+: EU261 Article 7 kicks in : large fixed compensation payment per passenger. *The lower bound triggers EU261 Article 7 monetary compensation at 3 hours arrival delay, the airline owes €250/€400/€600*

Minutes 300+: *It becomes cheaper to cancel the flight than to continue operating it. EU261 compensation is already owed in full — cancellation adds no marginal Article 7 cost Crew duty time limits are likely breached — forcing an unscheduled crew swap Overnight hotel + transfer costs become unavoidable for many passengers The downstream cascade has fully developed — 4-6 other flights are also delayed*

AIRCRAFT SIZES

SIZE	Distance in km	Num Passengers
S	< 1,500 km	100
M	1,500–5,000 km	150

L	> 5,000 km	250
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VARIABLE COST PER MINUTE

Short delays (<30 min) - per-minute cost

SIZE	COST per MIN
S	(15-30) - WE USE 20
M	(40-60) - WE USE 50
L	(110– 180) WE USE 140

FIXED COSTS (EU261)

SMALL

<i>Bracket</i>	<i>Time Interval</i>	<i>Severity</i>	<i>EU261 Cost (EUR)</i>
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR
2	60 - 120 min	Moderate delay	500 EUR
3	120 - 180 min	Major delay	500 EUR
4	180 - 300 min	Severe delay	26,000 EUR
5	300+ min	Cancellation zone	26,500 EUR

<i>EU261 trigger reference for this aircraft size:</i>	
<i>Article 9 (meal/refreshment vouchers): triggered at 120 min</i>	<i>~EUR 5-15 per passenger</i>
<i>Article 7 (short-haul, less than 1,500 km): triggered at 180 min</i>	<i>EUR 250 per passenger</i>
<i>Total Article 7 exposure (100 pax)</i>	<i>EUR 25,000</i>

MEDIUM

Bracket	Time Interval	Severity	EU261 Cost (EUR)
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR
2	60 - 120 min	Moderate delay	750 EUR
3	120 - 180 min	Major delay	750 EUR
4	180 - 300 min	Severe delay	61,500 EUR
5	300+ min	Cancellation zone	62,250 EUR

EU261 trigger reference for this aircraft size:	
Article 9 (meal/refreshment vouchers): triggered at 120 min	~EUR 5-15 per passenger
Article 7 (medium-haul, 1,500-3,500 km): triggered at 180 min	EUR 400 per passenger
Total Article 7 exposure (150 pax)	EUR 60,000

LARGE

Bracket	Time Interval	Severity	EU261 Cost (EUR)
0	0 - 15 min	On time	0 EUR
1	15 - 60 min	Minor delay	0 EUR
2	60 - 120 min	Moderate delay	0 EUR
3	120 - 180 min	Major delay	0 EUR
4	180 - 300 min	Severe delay	152,500 EUR
5	300+ min	Cancellation zone	153,750 EUR

Using this data for *hourly* payments, convert all the Table 16.4 hourly rates to per-minute, and find averages for the 3 size classes

Table 16.4: Strategic delay costs per hour – base scenario



Aircraft type	At gate	Taxiing	En-route
A319	€962	€2,281	€4,062
B734	€1,069	€2,566	€4,145
B735	€1,248	€2,756	€4,906
B738	€2,043	€5,036	€8,576
B752	€274	€891	€1,069
B763	€404	€1,164	€1,509
B744	€641	€2,031	€3,802
A319	€712	€2,221	€3,908
A320	€606	€2,031	€3,504
A321	€1,199	€2,459	€4,336
AT43	€1,782	€5,892	€13,007
AT72	€856	€2,839	€5,001
DH8D	€1,556	€4,015	€7,401
E190	€641	€1,354	€1,936
A332	€915	€2,043	€3,267

Then the final per minute rates are

Size	B1	B2	B3	B4
S	20	36	45	54
M	50	55	70	85
L	140	150	190	230

FINAL PIECEWISE FUNCTIONS (variable + fixed costs):

$$C_{small}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 20t & \text{if } 15 \leq t < 60 \\ 1,200 + 36(t - 60) & \text{if } 60 \leq t < 120 \\ 3,360 + 1,000 + 45(t - 120) & \text{if } 120 \leq t < 180 \\ 7,060 + 25,000 + 54(t - 180) & \text{if } 180 \leq t < 300 \\ 16,640 & \text{if } t \geq 300 \end{cases}$$

$$C_{medium}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 50t & \text{if } 15 \leq t < 60 \\ 3,000 + 55(t - 60) & \text{if } 60 \leq t < 120 \\ 6,300 + 1,500 + 70(t - 120) & \text{if } 120 \leq t < 180 \\ 12,000 + 60,000 + 85(t - 180) & \text{if } 180 \leq t < 300 \\ 25,720 & \text{if } t \geq 300 \end{cases}$$

$$C_{large}(t) = \begin{cases} 0 & \text{if } 0 \leq t < 15 \\ 140t & \text{if } 15 \leq t < 60 \\ 8,400 + 150(t - 60) & \text{if } 60 \leq t < 120 \\ 17,400 + 190(t - 120) & \text{if } 120 \leq t < 240 \\ 40,200 + 153,750 + 230(t - 240) & \text{if } 240 \leq t < 300 \\ 123,900 & \text{if } t \geq 300 \end{cases}$$